ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** curiosity	A curious mind keeps asking why and how.
2	**(a)** biodiversity	Biodiversity means diversity of life.
3	**(b)** carbohydrates	Carbohydrates give quick energy.
4	**(b)** magnetic	Magnetic materials include iron, nickel and cobalt.
5	**(d)** metre	SI unit of length is metre (m).
6	**(a)** material	An object is made of material.
7	**(a)** temperature	Temperature measures hotness.
8	**(a)** three	Solid, liquid and gas.
9	**(b)** handpicking	Used when pieces are large.
10	**(c)** they need food	All living beings need food for energy.
11	**(b)** natural resources	Air, water, soil are natural resources.
12	**(d)** celestial bodies	Bodies beyond the Earth.
13	**(c)** a systematic way of understanding nature	Science studies nature through observation and experiment.
14	**(c)** parallel venation	Monocot leaves have parallel veins.
15	**(a)** calcium	Calcium builds bones.
16	**(c)** small magnet	It is a tiny magnet.
17	**(d)** thread	A thread follows the curve.
18	**(d)** sugar	Sugar dissolves.
19	**(b)** mercury	Mercury expands uniformly.
20	**(c)** temperature is higher	Heat helps.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Repetition checks that the result is reliable and not due to chance (1); It reduces the effect of errors (1).	1+1
22	The variety of living organisms on Earth (1); It includes plants, animals and microorganisms (1).	1+1
23	Substances in food needed by the body (1); Examples: proteins and vitamins (1).	1+1
24	Magnetic: iron, nickel, cobalt (1); Non-magnetic: wood, plastic, glass (1).	1+1
25	Everyone gets the same value (1); Results can be compared and shared (1).	1+1
26	The substance an object is made of (1); Wood and iron (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Observe and ask a question (1); Make a hypothesis and test it by experiment (1); Record results and draw a conclusion (1).	1+1+1
28	Tomato: herb (1); Rose: shrub (1); Banyan: tree (1).	1+1+1
29	Rice: carbohydrate for energy (1); Pulses: protein for growth (1); Butter: fat for energy and warmth (1).	1+1+1
30	Move a magnet over the mixture (1); Iron nails stick to the magnet (1); Wooden pieces remain (1).	1+1+1
31	Place the zero of the ruler at one end (1); Keep your eye above the other end (1); Read the mark and write with unit (1).	1+1+1
	OR: Clear glass: transparent (1); Butter paper: translucent (1); Wooden door: opaque (1).	1+1+1
32	Wash and shake it below 35 °C (1); Place under the tongue or armpit for a minute (1); Read with the eye level to the mercury thread (1).	1+1+1
	OR: Sun heats water bodies (1); Vapour rises and cools (1); It condenses into tiny droplets forming clouds (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Observation: dry seeds do not sprout (1); Hypothesis: seeds need water (1); Experiment: one dish with water, one dry, same seeds and place (1); Result: only wet seeds sprout (1); Conclusion: seeds need water for germination (1).	1+1+1+1+1
34	Herbs: soft green stem, e.g. spinach (1); Shrubs: hard branching near the base, e.g. hibiscus (1); Trees: thick trunk, e.g. mango (1); Climbers: need support, e.g. pea (1); Creepers: spread on ground, e.g. pumpkin (1).	1+1+1+1+1
35	Carbohydrates: energy, rice (1); Proteins: growth, pulses (1); Fats: energy, butter (1); Vitamins: protection, fruits (1); Minerals: bones and blood, milk and spinach (1).	1+1+1+1+1
36	Collect various objects (1); Bring a magnet near each (1); Magnetic: iron nail, steel pin (1); Non-magnetic: wood, plastic, glass (1); Record in a table (1).	1+1+1+1+1
	OR: Linear: a car, a falling stone (1); Circular: stone whirled on a string, hands of a clock (1); Rotational: top, fan (1); Periodic: pendulum, swing (1); Random: butterfly (1).	1+1+1+1+1
37	Appearance: lustre of gold (1); Hardness: diamond (1); Solubility: sugar (1); Transparency: glass (1); Floating: cork, sinking: stone (1).	1+1+1+1+1
	OR: Glass tube with mercury bulb (1); Mercury expands on heating (1); Kink keeps the level (1); Shake before use (1); Do not hold by the bulb (1).	1+1+1+1+1
38	Sun heats sea (1); Evaporation (1); Condensation to clouds (1); Precipitation (1); Water flows back to rivers and seas (1).	1+1+1+1+1
	OR: Filter the mixture (1); Sand remains as residue (1); Filtrate has salt (1); Evaporate the filtrate (1); Salt crystals remain (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** condensation	Cooling vapour.
2	**(d)** reproduction	Offspring are formed.
3	**(d)** mixture of gases	Contains N2, O2 and others.
4	**(a)** Mercury	Smallest planet.
5	**(d)** a mobile phone	Technology is the practical use of science.
6	**(a)** herbivores	Cows are herbivores.
7	**(d)** carbohydrates	They contain starch.
8	**(a)** to keep them from losing magnetism	Keepers preserve magnetism.
9	**(b)** changes with time	Change of position.
10	**(c)** copper	Metals conduct heat well.
11	**(c)** conductors	Metals are good conductors.
12	**(b)** collecting and storing rainwater	Collect rain for use.
13	**(a)** distilled water	Only one substance.
14	**(c)** food, water and air	Basic needs.
15	**(b)** crude oil	Obtained from under the ground.
16	**(c)** new-moon day	Amavasya.
17	**(d)** observation	Observation is the first step of enquiry.
18	**(b)** habitat	A habitat provides food and shelter.
19	**(c)** proteins	Proteins build and repair tissues.
20	**(d)** iron nail	Iron is magnetic.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	A measure of how hot or cold a body is (1); It is measured with a thermometer (1).	1+1
22	Solid: ice (1); Liquid: water; gas: water vapour (1).	1+1
23	Two or more substances mixed together (1); Salt and sand (1).	1+1
24	Need food and respire (1); Grow, respond, reproduce (1).	1+1
25	Materials from nature that we use (1); Air and water (1).	1+1
26	Natural object in space (1); Sun and Moon (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Allow to stand so sand settles (1); Pour off water (decantation) (1); Filter the rest (1).	1+1+1
28	It does not need food (1); It does not grow or reproduce (1); It does not respond (1).	1+1+1
29	Nitrogen about 78 per cent (1); Oxygen about 21 per cent (1); Others about 1 per cent (1).	1+1+1
30	Moon has no light of its own (1); It reflects sunlight (1); It revolves around Earth (1).	1+1+1
31	Observation: what we see, e.g. the ground is wet (1); Inference: what we conclude, e.g. it has rained (1); Inference should be supported by evidence (1).	1+1+1
	OR: Goat: herbivore (1); Tiger: carnivore (1); Bear: omnivore (1).	1+1+1
32	No single food has all nutrients (1); Each nutrient has a different job (1); Variety makes a balanced diet (1).	1+1+1
	OR: Keep in pairs with keepers (1); Avoid heating and hammering (1); Keep away from electronic gadgets (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Need food (1); Respire (1); Respond to stimuli (1); Grow and reproduce (1); Excrete and die (1).	1+1+1+1+1
34	Nitrogen 78 per cent (1); Oxygen 21 per cent (1); Carbon dioxide small (1); For breathing (1); For burning and photosynthesis (1).	1+1+1+1+1
35	Sun at the centre (1); Mercury: nearest to the Sun (1); Jupiter: largest (1); Saturn: rings (1); Mars: red (1).	1+1+1+1+1
36	Physics: matter, energy and forces (1); Chemistry: substances and changes (1); Biology: living things (1); Astronomy: space objects (1); Geology: Earth and rocks (1).	1+1+1+1+1
	OR: Blade or lamina is the broad part (1); Petiole is the stalk (1); Midrib is the main vein (1); Veins carry water and food (1); Pattern of veins is venation (1).	1+1+1+1+1
37	Eat without distraction (1); Chew well (1); Choose healthy food (1); Do not waste (1); Eat at fixed times (1).	1+1+1+1+1
	OR: Magnetised needle on a pivot (1); Dial with directions (1); Hold it flat (1); Needle points north (1); Rotate the dial to match N (1).	1+1+1+1+1
38	Measure 100 sheets together (1); Total thickness is 1.2 cm (1); Divide by 100 (1); Gives 0.012 cm (1); Convert to 0.12 mm (1).	1+1+1+1+1
	OR: Place objects in water (1); Wood, cork float (1); Stone, nails sink (1); Float if lighter than water (1); Record observation (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–12) | Class 6 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 100 centimetres	1 m = 100 cm.
2	**(b)** transparent	We can see clearly through them.
3	**(b)** thermometer	Thermometer measures temperature.
4	**(c)** water vapour	Vapour is invisible water gas.
5	**(d)** winnowing	Wind blows away light husk.
6	**(d)** a stone	Stones do not grow or respire.
7	**(c)** renewable	E.g. sunlight, wind.
8	**(d)** star	It produces its own light and heat.
9	**(b)** a mango tree	Plants grow, respire and reproduce.
10	**(a)** tap root system	Mustard and carrot have tap roots.
11	**(a)** calcium	Milk contains calcium and protein.
12	**(b)** north-south	It aligns with Earth's field.
13	**(c)** directly above the mark	Avoids parallax error.
14	**(d)** sand	Sand does not dissolve.
15	**(b)** prevents mercury from falling back on its own	It keeps the reading.
16	**(b)** the surface area is larger	More surface is exposed.
17	**(c)** filtration	Sand is insoluble.
18	**(c)** growth	Living things grow.
19	**(d)** smoke from vehicles	Smoke contains harmful gases.
20	**(a)** Jupiter	Jupiter.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Science: understanding how nature works (1); Technology: using that knowledge to make useful things (1).	1+1
22	Tap root: carrot, mustard (1); Fibrous: wheat, grass (1).	1+1
23	Diet with all nutrients in proper amounts (1); It maintains health (1).	1+1
24	It is influenced by the Earth's magnetism (1); The Earth behaves like a large magnet (1).	1+1
25	Lay a thread along the path (1); Straighten the thread and measure it (1).	1+1
26	They are poor conductors of heat (1); So hands do not get burnt (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Zero may be missing (1); It gives wrong length (1); Use a clear full mark (1).	1+1+1
28	Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred fast (1).	1+1+1
29	Cloth is a poor conductor (1); Prevents burns (1); Heat is not transferred to the hand (1).	1+1+1
30	Ice melts to water at 0 °C (1); Water heats up to 100 °C (1); It boils to steam (1).	1+1+1
31	Boil or cool milk (1); Cream forms a layer (1); Remove it or churn (1).	1+1+1
	OR: Some seeds appear dry and inactive (1); Dry seeds sprout later (1); They show life only in suitable conditions (1).	1+1+1
32	Give oxygen (1); Hold soil (1); Give shade and food (1).	1+1+1
	OR: Shows north (1); Remains nearly fixed (1); Helps to find direction at night (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	One hand in hot water, one in cold (1); Then both in lukewarm water (1); Hot hand feels cold (1); Cold hand feels warm (1); Therefore use a thermometer (1).	1+1+1+1+1
34	Drinking, cooking (1); Farming (1); Close taps (1); Fix leaks (1); Harvest rain (1).	1+1+1+1+1
35	Heat sea water (1); Water evaporates (1); Condense vapour on a cool surface (1); Collect drops (1); Salt remains (1).	1+1+1+1+1
36	Light (1); Touch (1); Water (1); Roots grow downward (1); Shoots grow upward (1).	1+1+1+1+1
	OR: Weathering of rocks (1); Humus from decay (1); Topsoil (1); Subsoil (1); Bedrock (1).	1+1+1+1+1
37	Planets orbit the Sun (1); Moon orbits Earth (1); Orbit is a path (1); Held by gravity (1); Revolution takes different times (1).	1+1+1+1+1
	OR: Science is the study of nature through observation and tests (1); Technology uses scientific knowledge to make useful things (1); Science: laws of motion, properties of light (1); Technology: bicycle, telescope (1); Both depend on each other (1).	1+1+1+1+1
38	Tap root: one main root, e.g. carrot (1); Fibrous: many similar roots, e.g. wheat (1); Reticulate venation: net-like, e.g. peepal (1); Parallel venation: veins parallel, e.g. grass (1); Roots of grass are fibrous, matching parallel veins (1).	1+1+1+1+1
	OR: Scurvy: bleeding gums (1); Rickets: bent legs (1); Night blindness: poor dim-light vision (1); Anaemia: tiredness (1); Goitre: swelling in the neck (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND